

Bonsai | Project progress

30 September 2026 | Model-in-Silicon: Bonsai 2 27B ternary target

Foundation delivered; first executable slice validated in simulation. Week 0 is complete, Week 1 is active, and Weeks 2-6 remain backlog. Custom Bonsai FPGA results and full-model token generation are not yet demonstrated.

EVIDENCE DELIVERED

Week 0 / PR #1 merged. Interface and weight-store contracts, image tooling and reproducible environment are delivered. The existing baseline passed **1,332 selected targets**; vendor shell XSIM passed **9/9**. The official AWS F2 example executed **2,264 additions** on physical hardware. That validates the FPGA environment, not Bonsai model execution.

Week 1 / PR #2 draft. Model image -> sealed DDR/store -> DOT128 -> mailbox is implemented. The latest local workflow passed **16 steps: 396 native + 6 synthetic cases**, including two expected invalid-code rejections; both full-image readbacks matched. The public SDK transport passed **41 mock checks**. CodeQL passed on validated source 3ee0f99e; these are software/simulator results.

MILESTONE SCHEDULE

Milestone	Status	Completion gate / planned outcome
Week 0 - Baseline and interfaces	DONE	PR #1 merged; report and actual vendor FPGA evidence
Week 1 - Core RTL and tooling	ACTIVE	PR #2: custom XSIM, DCP/timing, AFI and returned-output tests
Week 2 - Ternary engine and fit	BACKLOG	Correctness, synthesis and measured FPGA fit
Week 3 - Helpers and intrinsics	BACKLOG	Integrated helpers, kernels and layer-level tests
Week 4 - Timing and F2 fit	BACKLOG	Named configurations with routed timing/resources
Week 5 - Attention and eight tiles	BACKLOG	Attention, eight-engine feasibility and inference candidate
Week 6 - Scheduling and sign-off	BACKLOG	Scheduling, end-to-end evaluation and release evidence

Week 1 planned due: 2 October 2026. This is the issue's planned date, not a completion forecast. Week labels express the milestone sequence.

NEXT THREE DELIVERABLES

- 1. Complete the Week 1 F2 gate:** execution is authorized; AWS browser login is pending and the target has not yet started. Run custom XSIM, implementation/timing and physical image/returned-output checks, then review and merge PR #2.
- 2. Attach the current evidence report:** publish this project snapshot and refresh the Week 1 report with exact source, simulator and actual FPGA artifacts as they become available.
- 3. Ternary-array design v0:** document dataflow, interfaces, numerics and resource/bandwidth assumptions. This is design work only; no new array RTL or fit claim.

PRINCIPAL RISKS AND BOUNDARIES

The current **Bonsai 1.7B Q1_0 fixture is binary sign quantization**; it does not validate the separate **27B ternary** target. Custom timing closure, memory bandwidth and eight physical engines remain unproven. Full Arcilator support and model integration remain follow-on work. Accountable owner: **Rachit Tibrewal**; Codex implements and runs checks with separate automated technical review. Each milestone closes with a reviewed/merged PR, runnable simulator, actual FPGA evidence for its scope and an attached report.

SOURCES

Week 0 report / PR #1 | Week 1 report / PR #2 | Project plan
Status reflects verified evidence through this report date; new F2 execution is pending.